

DEMAND SENSITIVITY ANALYSIS: ELASTICITY

$$\text{ELASTICITY} = \frac{\% \Delta \text{ IN } Y}{\% \Delta \text{ IN } X}$$

Y = DEPENDENT VARIABLE

X = INDEPENDENT VARIABLE

POINT PRICE ELASTICITY OF DEMAND

$$= \frac{\% \Delta \text{ IN QUANTITY DEMANDED OF } X}{\% \Delta \text{ IN PRICE OF } X}$$

ARC PRICE ELASTICITY OF DEMAND,
ถ้าเมื่อราคาสินค้าเปลี่ยนแปลงแล้ว 5 %.

ARC PRICE ELASTICITY OF DEMAND

$$= \frac{Q_2 - Q_1}{P_2 - P_1} \times \frac{P_2 + P_1}{Q_2 + Q_1}$$

Q = QUANTITY DEMANDED OF X

P = PRICE OF X.

DEMAND SENSITIVITY ANALYSIS: ELASTICITY

$$\text{ELASTICITY} = \frac{\text{PERCENTAGE CHANGE } Y}{\text{PERCENTAGE CHANGE } X}$$

Y = DEPENDENT VARIABLE

X = INDEPENDENT VARIABLE

$$\begin{aligned} \text{POINT PRICE ELASTICITY OF DEMAND} \\ = \frac{\% \Delta \text{QUANTITY OF } X}{\% \Delta \text{ IN PRICE OF } X} \end{aligned}$$

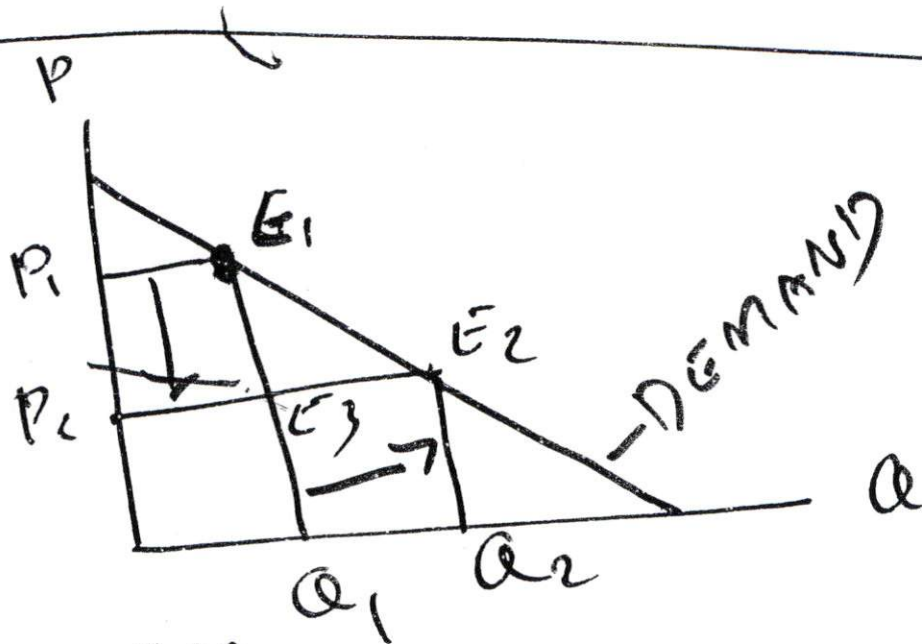
$$\begin{aligned} \text{ARC PRICE ELASTICITY OF DEMAND} \\ = \frac{\% \Delta Q_x}{\text{LARGE } \% \Delta \text{ in } P_x \text{ (MORE THAN 5\%)}} \end{aligned}$$

$$|E_P| < 1 = \text{INELASTIC}$$

$$|E_P| = 1 = \text{UNITARY ELASTIC}$$

$$|E_P| > 1 = \text{ELASTIC}$$

POINT PRICE ELASTICITY OF DEMAND



$$E_p = \frac{\gamma \cdot \Delta Q}{\gamma \cdot \Delta P}$$

$$= \frac{\frac{Q_2 - Q_1}{Q_1} \times 100}{\frac{P_2 - P_1}{P_1} \times 100}$$

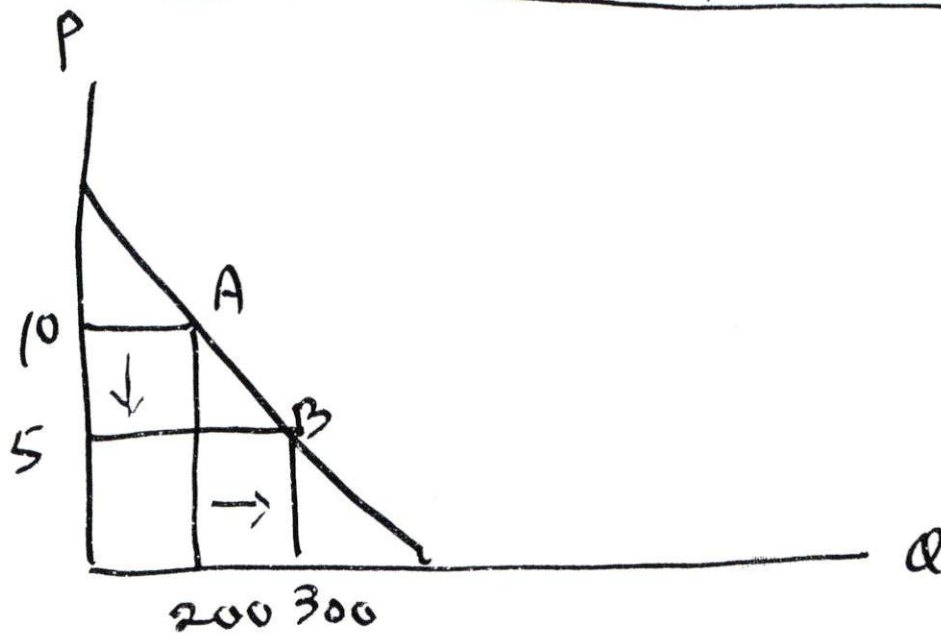
$$= \frac{Q_2 - Q_1}{P_2 - P_1} \times \frac{P_1}{Q_1}$$

$$= \frac{\Delta Q}{\Delta P} \times \frac{P_1}{Q_1}$$

$$= \frac{dQ}{dP} \cdot \frac{P_1}{Q_1}$$

POINT PRICE ELASTICITY OF DEMAND

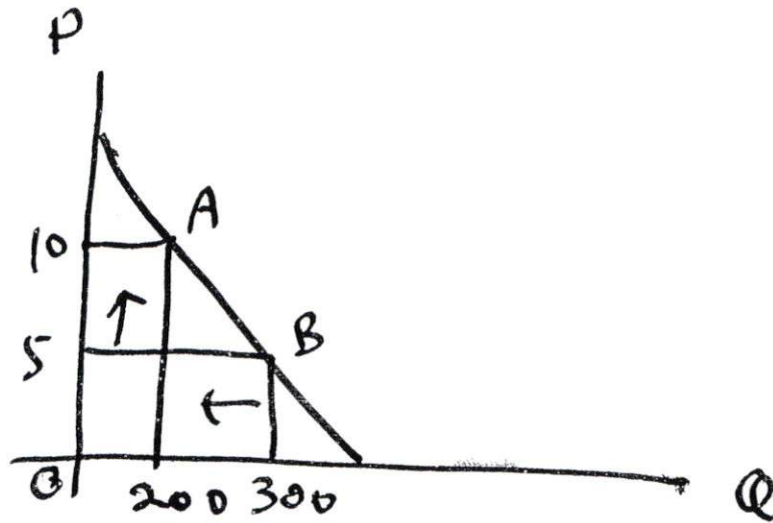
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2. Point ELASTICITY OF DEMAND ^{PRICE} at point A

$$\begin{aligned} E_p &= \frac{\Delta Q}{\Delta P} \times \frac{P}{Q} \\ &= \frac{300-200}{5-10} \times \frac{10}{200} \\ &= \frac{100}{-5} \times \frac{10}{200} \\ &= -1 \end{aligned}$$

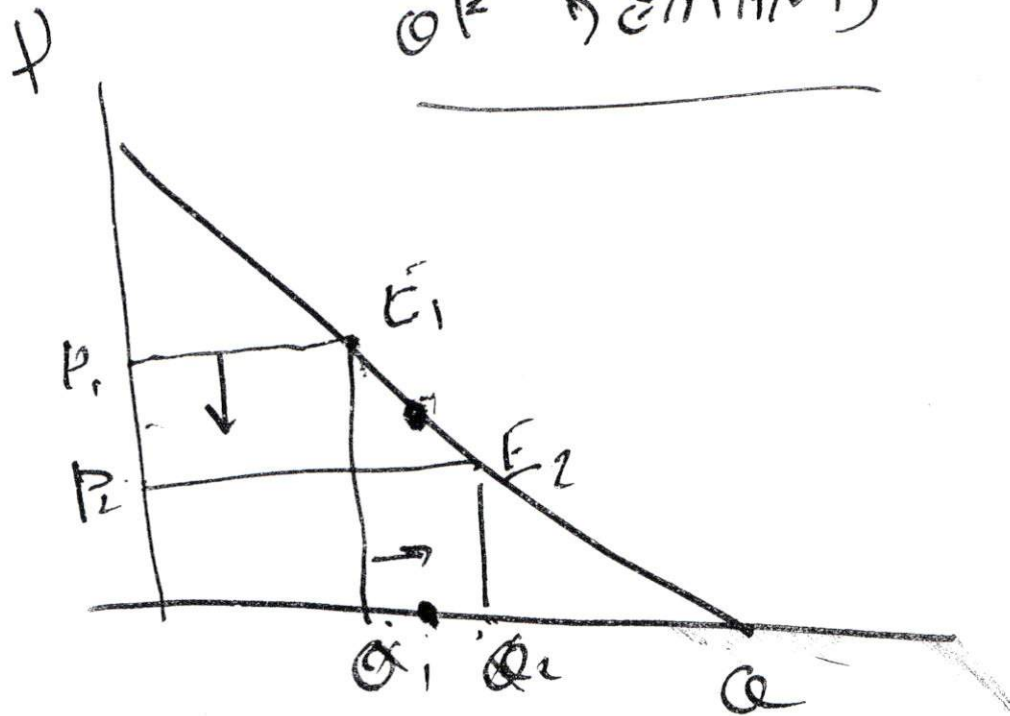
POINT PRICE ELASTICITY OF DEMAND (2)



∴ POINT PRICE ELASTICITY AT POINT B.

$$\begin{aligned} E_p &= \frac{\Delta Q}{\Delta P} \times \frac{P}{Q} \\ &= \frac{200 - 300}{10 - 5} \times \frac{5}{300} \\ &= \frac{-100}{5} \times \frac{5}{300} \\ &= -\frac{1}{3} = 0.33 \end{aligned}$$

ARE PRICE ELASTICITY OF DEMAND



$$E_p = \frac{\% \Delta Q}{\% \Delta P}$$

$$= \frac{\frac{Q_2 - Q_1}{\frac{Q_2 + Q_1}{2}} \times 100}{\frac{P_2 - P_1}{\frac{P_2 + P_1}{2}} \times 100}$$

$$= \frac{Q_2 - Q_1}{\frac{Q_2 + Q_1}{2}} \times \frac{P_2 + P_1}{P_2 - P_1} \times 100$$

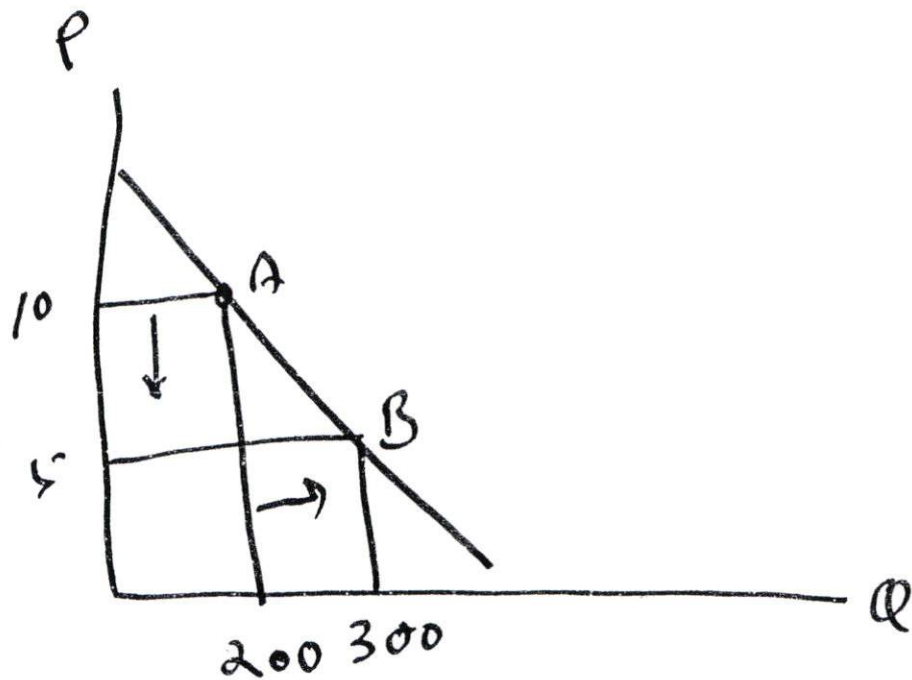
$$= \frac{Q_2 - Q_1}{Q_2 + Q_1} \times \frac{P_2 + P_1}{P_2 - P_1}$$

$$= \frac{Q_2 - Q_1}{P_2 - P_1} \cdot \frac{P_2 + P_1}{Q_2 + Q_1}$$

$$= \frac{\Delta Q}{\Delta P} \cdot \frac{P_2 + P_1}{Q_2 + Q_1}$$

$$= \frac{dQ}{dP} \cdot \frac{P_2 + P_1}{Q_2 + Q_1}$$

ARC PRICE ELASTICITY OF DEMAND (3)



ARC PRICE ELASTICITY OF DEMAND.

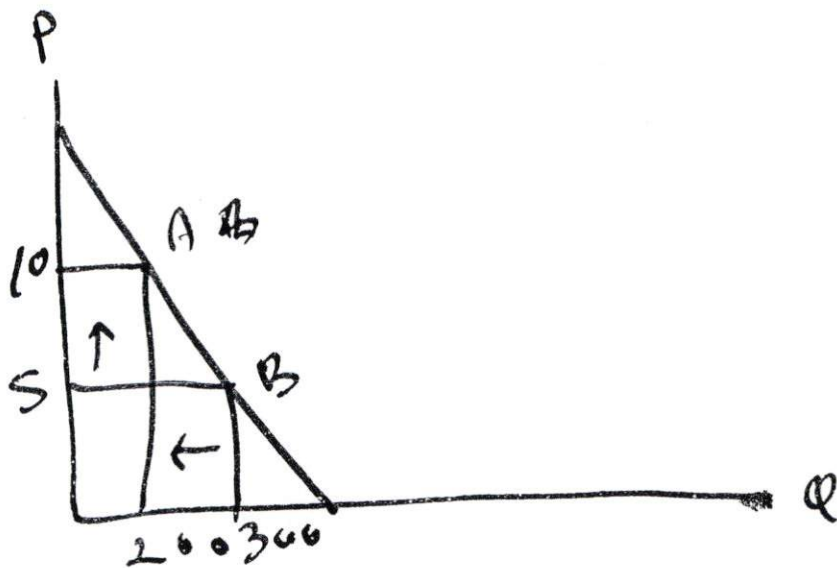
$$E_p = \frac{300 - 200}{5 - 10} \times \frac{10 + 5}{300 + 200}$$

$$= -\frac{100}{5} \times \frac{15}{500}$$

$$= -\frac{3}{5} = -0.6$$

ARC PRICE ELASTICITY OF DEMAND

(4)



Wm ARC PRICE ELASTICITY in q a B

$$\begin{aligned} E_p &= \frac{200-300}{10-5} \times \frac{5+10}{300+200} \\ &= \frac{-100}{5} \times \frac{15}{500} \\ &= -\frac{3}{5} = -0.6 \end{aligned}$$

គណនីបញ្ចេញប្រាក់ ១០០ លានរៀល ១០០ លានរៀល
បញ្ចេញប្រាក់

- 1 និយត្តិទំនើប Ep តាំ
- 2 និយត្តិប្រាក់បៀវត្ស Ep តាំ
- 3 និយត្តិក៏ដូចនិយត្តិទំនើប ឆ្នាំរៀល
បាតប្រាក់ Ep តាំ
- 4 និយត្តិក៏ដូចនិយត្តិទំនើប បាតប្រាក់
៧៩/១០ ឆ្នាំរៀល បាតប្រាក់ Ep តាំ
- 5 និយត្តិក៏ដូចនិយត្តិទំនើប បាតប្រាក់
បាតប្រាក់, Ep តាំ
- 6 និយត្តិបញ្ចេញប្រាក់ ឆ្នាំរៀល បាតប្រាក់
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DP	$ E < 1$	$ E = 1$	$ E > 1$
$P \uparrow$	$TR \uparrow$	$\bar{TR}$	$TR \downarrow$
$P \downarrow$	$TR \downarrow$	$\bar{TR}$	$TR \uparrow$

กรณี $|E_p| < 1$ เราต้องมองที่ $\pi \uparrow$
 มอง: $\uparrow P$ หรือ $\downarrow P$

$$\pi = TR - TC$$

ดังนั้น

$$\uparrow P \rightarrow TR \uparrow \quad \text{หรือ} \quad TC \downarrow$$

$$\therefore \pi \uparrow$$

$$\text{กรณี } |E_p| > 1$$

$$\downarrow P \rightarrow TR \uparrow \quad \text{หรือ} \quad \text{ต้องดูที่ } TC \uparrow$$

$\therefore \pi$ จะเพิ่มขึ้นหรือจะลดลงขึ้นอยู่กับ

TR หรือ TC ~~อะไร~~ ~~จะ~~ ~~ขึ้น~~ ~~กับ~~

INCOME ELASTICITY OF DEMAND (E_y)

$$E_y = \frac{\% \Delta Q_d}{\% \Delta Y}$$

if $E_y > 1$ = SUPERIOR GOODS (also
CYCLICAL NORMAL GOODS)

if $0 \leq E_y < 1$ = NORMAL GOODS (also
NON-CYCLICAL NORMAL
GOODS)

if $E_y < 0$ = INFERIOR GOODS

CROSS-PRICE ELASTICITY OF DEMAND

$$E_{xy} = \frac{\% \Delta Q_x}{\% \Delta P_y}$$

ถ้า $E_{xy} < 0$, X และ Y เป็นสินค้าที่
ใช้ร่วมกัน (COMPLEMENTARY
GOODS)

ถ้า $E_{xy} > 0$, X และ Y เป็นสินค้าที่
ทดแทนกันได้ (SUBSTITUTION GOODS)

ถ้า $E_{xy} = 0$, X และ Y เป็นสินค้าที่
ไม่เกี่ยวข้องกัน
(INDEPENDENT GOODS)

ความสัมพันธ์ระหว่าง DEMAND (P) MARGINAL REVENUE

ให้ $P = a - bQ$ เป็นสมการของ DEMAND
ซึ่งเส้นตรง

a = intercept ของเส้น Demand

b = SLOPE ของเส้น Demand

สมการของเส้น Total Revenue คือ

$$\begin{aligned} TR &= P \cdot Q \\ &= (a - bQ) \times Q \end{aligned}$$

$$TR = aQ - bQ^2$$

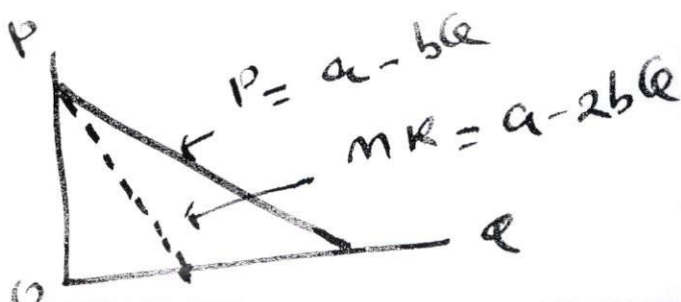
$$MR = \frac{dTR}{dQ} = a - 2bQ$$

a = intercept ของเส้น MR

$2b$ = Slope ของเส้น MR

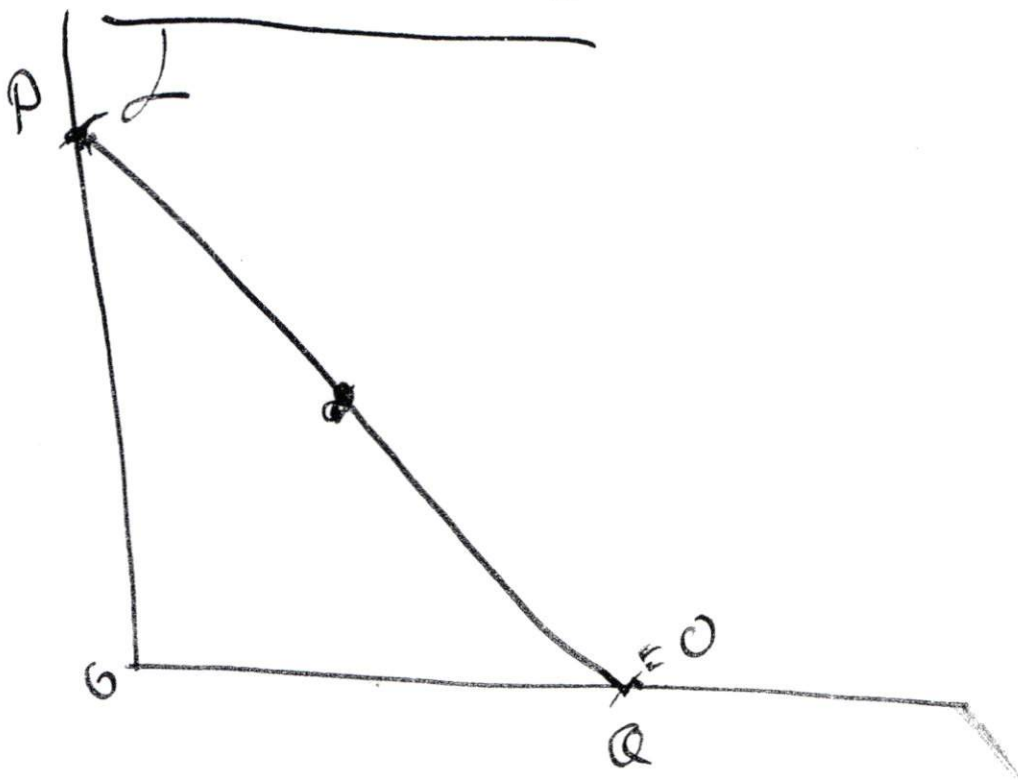
$\therefore$ SLOPE ของเส้น MR คือ 2 เท่า

ของเส้น DEMAND P



POINT PRICE ELASTICITY FOR DEMAND

အကျဉ်းချုပ်,



$$E_p = \frac{\Delta Q}{\Delta P} \cdot \frac{P}{Q}$$

$$\frac{\Delta Q}{\Delta P} = \text{CONSTANT (ဘာ: } \checkmark \text{ DEMAND)}$$

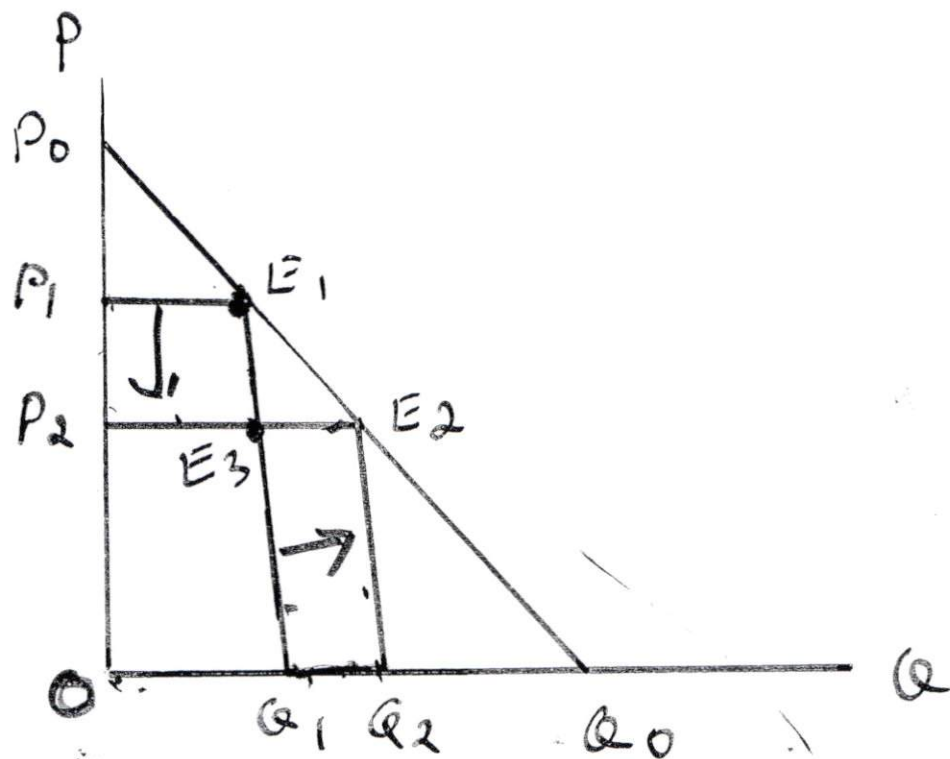
အကျဉ်းချုပ်,

$$\text{at } P, Q = 0 \therefore E_p = \infty$$

$$\text{at } Q, P = 0 \therefore E_p = 0$$

POINT PRICE ELASTICITY OF DEMAND

หาจุดใด ๆ บนเส้น DEMAND กราฟที่
เส้น DEMAND เส้นตรง



POINT PRICE ELASTICITY หาจุด E_1

$$\begin{aligned}
 E_{P_1} &= \frac{\Delta Q}{\Delta P} \times \frac{P}{Q} \\
 &= \frac{Q_2 - Q_1}{P_2 - P_1} \times \frac{OP_1}{OQ_1} \\
 &= \frac{E_2 E_3}{E_1 E_3} \times \frac{OP_1}{OQ_1} \\
 &= \frac{Q_0 Q_1}{Q_1 E_1} \times \frac{OP_1}{OQ_1}
 \end{aligned}$$

$$\frac{\Delta Q}{Q} \div \frac{\Delta P}{P} = E_P$$

$$\therefore E_P = \frac{\Delta Q}{Q} \div \frac{\Delta P}{P}$$

$$= \frac{\Delta Q \cdot P}{Q \cdot \Delta P}$$

$\therefore$ ถ้า E_P เป็นค่าที่น้อยกว่า 1 แสดงว่า DEMAND POINT PRICE ELASTICITY น้อยกว่า 1 หรือ $E_P < 1$ แสดงว่า = 1 หรือ

PRICE, TR, "a": MR

$$TR = P \cdot Q$$

$$MR = \frac{dTR}{dQ} = P \cdot \frac{dQ}{dQ} + Q \frac{dP}{dQ}$$

$$= P + Q \frac{dP}{dQ}$$

$$= P \left(1 + \frac{Q}{P} \cdot \frac{dP}{dQ} \right)$$

$$\therefore MR = P \left(1 + \frac{1}{E_p} \right)$$

FIRM MAXIMIZE PROFIT $\therefore MC = MR$

(સમગ્રજીવ Mc ને slope બેઠક +)

$$\therefore MC = P \cdot (1 + \frac{1}{\epsilon_p})$$

$$\rho = \frac{mc}{1 + \frac{1}{\epsilon_p}}$$

ร้านเดี๋ยวกีฬาลดราคาลง 5% ส่งผลให้

ปริมาณขายเพิ่มขึ้นจาก 32 หน่วย เป็น

36 หน่วย ราคาเดิมราคาหน่วยละ 16 บาท

ผู้ขายต้องการกำไรสูงสุด จะตั้งราคาเท่า

ไร ๆ กำไรเฉลี่ย $MC = 10$ บาท

วิธีที่

$$E_p = \frac{\Delta Q}{\Delta P} \times \frac{P}{Q}$$

$$= \frac{4}{-0.8} \times \frac{16}{32}$$

$$= -2.5$$

P

$$= \frac{MC}{1 + \frac{1}{E_p}}$$

$$= \frac{10}{1 + \frac{1}{E_p}}$$

$$= \frac{10}{1 + \frac{1}{-2.5}}$$

$$= 10$$

$$= \frac{10}{1 - 0.4}$$

$$= 16.67$$